



AMRITA INSTITUTE OF MEDICAL SCIENCES & RESEARCH CENTRE
PAN NO: AAATM2403M , Appointment call: 01292851234
Mata Amritanandamayi Marg, Sector 88, Faridabad, Haryana 121002

MRD No: 112145	Gender: Male	Age: 66Y 10M 15D
Name: VIPAN KUMAR GARG	Visit Type: OP	Visit Code: OP0002
Service: PET-CT Whole Body - CECT		

RESULTREPORT

Findings

18F-FDG WHOLE BODY PET-CECT

CLINICAL HISTORY: Case of multiple myeloma ISS II, RISS II. Diagnosed in June 2022, Baseline PET/CT (13.6.22). Post 6cycles of chemotherapy, PET/CT (9.12.22): metabolic resolution of the osseus lesions. On maintenance chemotherapy till Aug 23. FDG PET/CT (at PGI, 10.8.23): no significant interval change noted. During follow-up increase in kappa FLC, suspicion for recurrence. PET/CT for further evaluation.

PROTOCOL: Whole body FDG PET CT scan was performed from vertex to feet on Siemens Biograph Horizon Time-of-Flight PET CT scanner following intravenous injection of 10mCi of 18F-FDG under standard precaution and patient preparations. Patient was asked to rest for 60min in a shielded room to allow tracer to accumulate in the body. High resolution CT scan was performed during this examination on a 16 slice MDCT with i.v injection of non-ionic contrast (low volume Visipaque, 1mL/Kg) followed by PET images. Additional brain protocol was performed.

The creatinine level of the patient was 1.2mg/dl and blood sugar level was 119mg/dl at the time of tracer injection.

FINDINGS:

Overall quality of the study - Good. SUVmax mentioned in the report is in g/ml standardized to lean body mass.

BRAIN: Mild age related cortical atrophic changes are noted in bilateral cerebral and cerebellar hemispheres. The brain parenchyma is unremarkable with normal FDG biodistribution. No significant focal lesion or abnormal focal FDG uptake noted.

[Note: All brain lesions may not be apparent on 18F-FDG PET/CT and MRI brain may be performed, if clinically indicated.]

HEAD AND NECK:

Non FDG avid mucosal thickening is seen in posterior wall of nasopharynx and nasal cavity, likely inflammatory.

Dental related artefacts are noted partially obscuring the loco-regional anatomy.

Mild FDG uptake is seen in bilateral palatine tonsils with few bilateral tonsilloliths and bilateral level IB & II cervical lymph nodes, likely infective / inflammatory.

Mild FDG uptake is seen in few subcentimetric right supraclavicular lymph nodes (measuring 0.9 x 0.7cm, SUVmax 2.3).

Normal physiologic FDG distribution is seen in rest of the head-neck region.

Visualized paranasal sinuses and skull base appear normal. The suprahyoid neck, infrahyoid neck, thyroid, vascular structures and thoracic inlet do not show any obvious abnormality.

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THORAX:

Non FDG avid subcentimetric bilateral axillary lymph nodes are noted, likely reactive etiology. Increased FDG uptake is seen in multiple irregular fibronodules, few arranged in tree-in-bud pattern with adjacent ground glass haziness in the left lung upper & lower lobes (SUVmax 3.7). Non FDG avid ground glass haziness is seen in right lung lower lobe. Non FDG avid subpleural atelectatic changes are seen in the right middle lobe and bilateral lung lower lobes. Non FDG avid tiny nodule is noted in the right lung upper lobe, superior basal segment of right lung lower lobe. Few non-FDG avid thin walled air cysts are seen in bilateral lung parenchyma. FDG avid heterogeneously enhancing enlarged lower paratracheal, subcarinal and bilateral hilar (on the right measuring 2.0 x 1.7cm, SUVmax 5.3) lymph nodes are noted, few of the lymph nodes show internal calcifications. Post surgery (hiatus hernia repair) status. There is trace pericardial effusion. Coronary artery calcifications are noted. Mild atheromatous calcifications are noted in the thoracic aorta, abdominal aorta & its branches. Physiologic FDG uptake is seen in the myocardium. Large airways are within normal limits.

ABDOMEN-PELVIS:

Hepatomegaly is noted (liver measures 17.7cm in cranio-caudal extent). Non FDG avid relative well defined hypodense lesion is seen in segment III (measuring 1.1 x 1.1cm), favours benign etiology. No obvious focal FDG avid lesion / bilobar IHBR dilatation noted. No definite contrast filling defect is seen in the portal and hepatic venous systems. Gall bladder is partially distended without any obvious abnormality. [*Note: USG is the modality of choice for evaluation of gall stones]. Visualized stomach is poorly distended. Mild bilateral perinephric fat stranding is noted. Prostate appears enlarged (measures 5.1 x 5.5cm, AP x TR). No obvious focal FDG uptake noted. Normal FDG distribution is noted in the liver, spleen, gastrointestinal tract, kidneys and urinary bladder. The spleen, kidneys, adrenals, bowel, urinary bladder and pancreas appear normal. Omentum and peritoneum show no abnormality. No ascites is noted. No structural abnormality noted in the abdominal wall. No obvious FDG avid abnormal lymphadenopathy noted in the abdomen-pelvis.

MUSCULOSKELETAL SYSTEM AND OTHERS:

Degenerative changes are seen in the spine. Sacralisation of L5 vertebra is noted. Mild FDG uptake is seen in periarticular regions of bilateral shoulder joints, likely inflammatory. Non FDG avid benign sclerotic bone islands are seen in the head of left femur, left pubic symphysis and left acetabulum. Increased FDG uptake is seen in pathological fractures involving: right 6th rib laterally (with soft tissue component, SUVmax 5.4) and left 8th rib posteriorly (SUVmax 2.9); mild focal FDG uptake at the right 3rd rib costo-chondral junction (SUVmax 2.0). Increased FDG uptake is seen in collapse compression involving D6 vertebra with evidence of paravertebral soft tissue component (SUVmax 4.5). Non FDG avid lytic lesions associated with vertebral collapse involving D1 and D2 is noted. Few other non FDG avid lytic lesions are noted in the following skeletal sites: sternum, head of right femur and multiple dorsal-lumbar-sacral vertebrae. Physiologic FDG distribution is seen in rest of the axial and appendicular skeleton.

Comments

IMPRESSION: Case of multiple myeloma, post chemotherapy with increase in kappa FLC during follow-up. PET/CT reveals:

1. Metabolically active lytic lesions involving few skeletal sites as described (pathological fractures involving right 6th & left 8th ribs, collapsed D6 vertebra) - residual / recurrent neoplastic disease involvement.

2. Non-metabolic lytic lesions in multiple other skeletal sites as described above; D1 & D2 vertebral collapse with no significant metabolic activity.

3. Also noted:

(i) Metabolically active multiple irregular fibronodules, few arranged in tree-in-bud pattern with adjacent ground glass haziness in the left lung with multiple mediastinal & right supraclavicular lymph nodes - favors active infective/inflammatory etiology.

(ii) Hepatomegaly.

(iii) Prostatomegaly.

4. No definite scan evidence of abnormal metabolically active disease in rest of the body in this study.

*Previous FDG PET/CT scan CD not available for comparison (done at PGI, Chandigarh, dated 10.08.2023).

(Please carry report and CD on your next visit for comparison).

N.B. : This is only a professional opinion and not the final diagnosis. Radiological investigations are subject to variations due to technical limitations. Hence, correlation with clinical findings and other investigations should be carried out to know true nature of illness.

Verified By: Dr. Meghana Prabhu S

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